# Scalable Evaluation of Online Facilitation Strategies via Synthetic Simulation of Discussions

**Anonymous submission**


## Abstract

Limited large-scale evaluations exist for facilitation strategies of online discussions due to significant costs associated with human involvement. An effective solution is synthetic discussion simulations using Large Language Models (LLMs) to create initial pilot experiments. We propose design principles based on existing methodologies for synthetic discussion generation. Based on these principles, we propose a simple, generalizable, LLM-driven methodology to prototype the development of LLM facilitators by generating synthetic data without human involvement, and which surpasses current baselines. We use our methodology to test whether current Social Science strategies for facilitation can improve the performance of LLM facilitators. We find that, while LLM facilitators significantly improve synthetic discussions, there is no evidence that the application of these strategies leads to further improvements in discussion quality. In an effort to aid research in the field of facilitation, we release a large, publicly available dataset containing LLM-generated and LLM-annotated discussions using multiple open-source models. This dataset can be used for LLM facilitator finetuning as well as behavioral analysis of current out-of-the-box LLMs in the task. We also release an open-source python framework that efficiently implements our methodology at great scale.


**Framework** —
https://anonymous.4open.science/r/framework-850F
**Replication Code** —
https://anonymous.4open.science/r/experiments-F54D
**Dataset** —
https://anonymous.4open.science/r/experiments-F54D/data/datasets/main/main.zip

## 1 Introduction

The modern social media environment has evolved to be extremely demanding, with users facing ever-increasing threats such as targeted misinformation (Clemons, Schreieck, and Waran 2025; Denniss and Lindberg 2025), hate speech (Kolluri, Murthy, and Vinton 2025), and polarization (Pranesh and Gupta 2024). These threats can cause serious emotional and mental harm (Schluger et al. 2022). They can also lead to radicalization (Cho et al. 2024) and real-world violence (Schaffner et al. 2024). Furthermore, these threats can sabotage democratic dialogue (Esau, Friess, and Eilders 2017; Falk et al. 2021; Seering 2020), undermine trust in democratic institutions (Schroeder, Roy, and Kabbara 2024), and diminish the quality of information (Amaury and Stefano 2022). Platform designers and researchers traditionally focused on flagging and removing problematic content ("content moderation"—Seering (2020); Cresci, Trujillo, and Fagni (2022)), but these methods are no longer sufficient in practice (Horta Ribeiro, Cheng, and West 2023; Schaffner et al. 2024; Small et al. 2023; Korre et al. 2025). However, online communities benefit when moderators actively discuss with users. This approach is known as "conversational moderation" or "facilitation" (Argyle et al. 2023; Korre et al. 2025; Falk et al. 2021)). Facilitation prevents problematic user behavior before it surfaces (Cho et al. 2024; Seering 2020; Cresci, Trujillo, and Fagni 2022; Amaury and Stefano 2022), and supports community deliberation and group decision-making (Kim et al. 2021; Seering 2020).

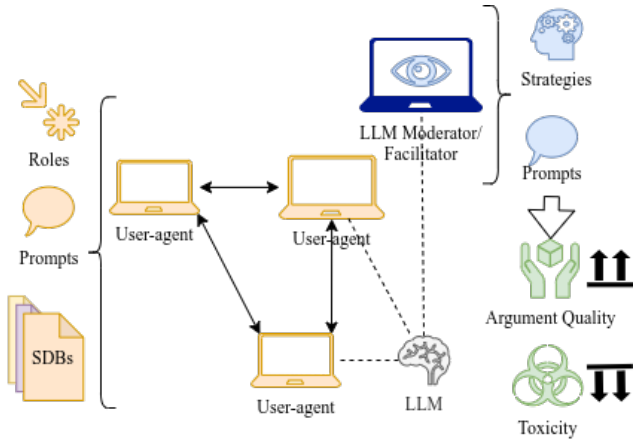



Figure 1: LLM user-agents with distinct SocioDemographic Backgrounds (SDBs—see §3.2) participate in a discussion, while the LLM moderator monitors and attempts to improve the quality of the discussion. We need to design prompts and configurations for both types of LLM agents.

Large Language Models (LLMs) have been hypothesized to be capable of facilitation tasks and can be scaled to a far greater extent compared to human facilitators (Korre et al. 2025; Small et al. 2023), making them a viable choice for

modern large-scale social networks. However, experimentation and development on these systems is hampered due to the costs of human participation—in this case, human discussants and evaluators (Rossi, Harrison, and Shklovski 2024). We posit that simulations with all-LLM-agents can be a cheap and fast way to develop and test LLM facilitators, initial versions of which may be unstable or unpredictable (Atil et al. 2025; Rossi, Harrison, and Shklovski 2024), before testing them with human participants.

Our work asks the following RQ: *How do we design and evaluate Synthetic Discussion Generation (SDG) methodologies satisfying registered criteria*? To answer this question we draw examples from methodologies proposed in literature, to establish basic design principles (§3.1), and propose a methodology enabling rapid model "debugging" (e.g., discarding suboptimal LLM prompts) and testing without human involvement (Fig. 1, §3.2). We validate the conformance to these principles through an ablation study (§5.2). To show the impact of our approach, we implement a framework based on the proposed methodology, which allows the design and evaluation of facilitation strategies proposed in modern Social Science research. We then investigate how they can enhance the performance of LLM facilitators and compare them with two common facilitation setups (§4.2). We find that while the presence of LLM facilitators has a *positive, statistically significant* influence on the quality of synthetic discussions, facilitation strategies inspired by Social Science research often *do not outperform simpler strategies* (§5.1). We also discover previously unreported aberrant behavior on the part of the LLM facilitator, in the form of excessive policing.

Finally, we release an open-source Python framework, available via PIP, that implements our methodology at scale, enabling the research community to rapidly experiment with LLM-based facilitators. Given that existing facilitation datasets are few and generally small (Korre et al. 2025), we also release a large, publicly available dataset with LLM-generated and annotated synthetic discussions (§6). Our dataset can be used for LLM facilitator finetuning (Ulmer et al. 2024), as well as for analyzing the behavior of out-of-the-box LLMs in the task of online facilitation. We use open-source LLMs and include all relevant configurations in order to make our study as reproducible as possible.

## 2 Background and Related Work

### 2.1 Synthetic Discussions

While studies exist for simulating user interactions in social media (Park et al. 2022; Mou, Wei, and Huang 2024; Törnberg et al. 2023; Rossetti et al. 2024), and for using LLM facilitators (Kim et al. 2021; Cho et al. 2024), none so far have combined the two approaches.

Balog et al. (2024) extract topics and comments from online human discussions and prompt an LLM to continue them. However, they do not use LLM-based user agents to simulate conversational dynamics, nor do they include facilitators in their setup. Additionally, their approach depends on the availability of human discussion datasets with the desired topics. Ulmer et al. (2024) create synthetic discussions between two roles: an agent controlling a fictional environment and a client interacting with it. These discussions are filtered and used to finetune the agent LLM for a specific task. Our methodology generalizes their framework: an agent (facilitator) interacts with multiple clients (non-facilitator users). Finally, Abdelnabi et al. (2024) generate synthetic negotiations involving multiple agents with different agendas and responsibilities. Our work can be seen as a domain shift of their approach—from negotiation to discussion facilitation—where various user types (e.g., normal users, trolls, community veterans) engage in discussion overseen by a facilitator with veto power.

### 2.2 LLM Facilitation

Unlike classification models traditionally used in online platforms, LLMs can actively facilitate discussions (Korre et al. 2025). They can warn users for rule violations (Kumar, AbuHashem, and Durumeric 2024), monitor engagement (Schroeder, Roy, and Kabbara 2024), aggregate diverse opinions (Small et al. 2023), and provide translations and writing tips—which is especially useful for marginalized groups (Tsai et al. 2024). These capabilities suggest that LLMs may be able to assist or even replace human facilitators in many tasks (Small et al. 2023; Korre et al. 2025).

Moderator chatbots have shown promise; Kim et al. (2021) demonstrated that simple rule-based models can enhance discussions, although their approach was largely confined to organizing the discussion based on the "think-pair-share" framework (Nik Ahmad 2010; Navajas, Niella, and Garbulsky 2018), and balancing user activity. Cho et al. (2024) use LLM facilitators in human discussions, with prompts based on Cognitive Behavioral Therapy and the work of Rosenberg and Chopra (2015). They show that LLM facilitators can provide high-quality feedback to users, although they struggle to make users more respectful and cooperative. In contrast to both works, ours uses exclusively LLM participants and LLM facilitators, and tests the latter in an explicitly toxic and challenging environment.

### 2.3 Discussion Quality

We need to evaluate two different quality dimensions. One is *discussion quality as seen by humans*, which is difficult to measure, both because of the breadth of the possible goals of a discussion, and because of the lack of established computational metrics in Social Science literature (Korre et al. 2025). There are however some that could reasonably be applied in this domain, such as toxicity (De Kock, Stafford, and Vlachos 2022; Xia et al. 2020), connective language (Lukito et al. 2024) and political discussion quality (Jaidka 2022).

The second quality dimension is measuring "high-quality" or "useful" data. This is essential in LLM-based discussion frameworks, as such discussions tend to deteriorate quickly without human involvement, often becoming repetitive and low-quality (Ulmer et al. 2024). Despite this importance, methods for quantifying the quality of synthetic data remain limited.

Balog et al. (2024) use a mix of graph-based, methodology-specific, and lexical similarity metrics, many of which depend on human discussion datasets. Their most

generalizable measure is a loosely defined "coherence" score, which is LLM-annotated without theoretical grounding. Kim et al. (2021) assess quality through post-discussion surveys and by measuring lexical diversity to approximate the variety of opinions expressed. Ulmer et al. (2024) introduce a discussion-level metric called *"Diversity"*, which penalizes repeated text sequences between comments using average pairwise ROUGE-L (Lin 2004) scores. Their approach suffers from the limitations of ROUGE scores (mainly the use of exact-word matching), but their metric is computationally efficient, explainable and independent from any specific domain and dataset.

### 2.4 LLMs as Human Subjects

While there is always a desire for synthetic simulation systems to be "realistic" w.r.t. human behavior (Grossmann et al. 2023; Törnberg et al. 2023; Argyle et al. 2023), this can not be claimed nor reliably measured by using LLM agents in lieu of humans (Rossi, Harrison, and Shklovski 2024).

It is true that LLMs have demonstrated complex, emergent social behaviors (Park et al. 2023a; Marzo, Pietronero, and Garcia 2023; Leng and Yuan 2024; Abdelnabi et al. 2024; Abramski et al. 2023; Hewitt et al. 2024; Park et al. 2024). However, significant limitations of LLMs remain in the context of Social Science experiments. Issues include undetectable behavioral hallucinations (Rossi, Harrison, and Shklovski 2024); sociodemographic, statistical and political biases (Anthis et al. 2025; Hewitt et al. 2024; Rossi, Harrison, and Shklovski 2024; Taubenfeld et al. 2024); unreliable survey responses (Jansen, gyo Jung, and Salminen 2023; Bisbee et al. 2024; Neumann, De-Arteaga, and Fazelpour 2025; Gligori'c et al. 2024); non-deterministic outputs (Atil et al. 2025; Bisbee et al. 2024); and excessive agreeableness (Park et al. 2023b; Anthis et al. 2025; Rossi, Harrison, and Shklovski 2024).

Thus, *no study based on LLM agents* can claim that it generates"realistic" discussions; reproduction studies with humans are ultimately needed.

## 3 Methodology

### 3.1 Designing synthetic discussions

Many SDG frameworks have been proposed in literature; both simple (Tsai, Qian, and comm. contributors 2025; Ulmer et al. 2024) and complex (Balog et al. 2024; Abdelnabi et al. 2024; Park et al. 2023a). What makes these frameworks widely used (perhaps modified) by other people has not been explored in literature, despite many such implementations. Concordia (Vezhnevets et al. 2023) is an example of a general, complex framework. While technically impressive, it has failed to garner widespread adoption despite efforts to promote it. This is evidenced by most recent publications creating their own SDG frameworks.

In the field of Software Engineering, there is a widely shared notion that simple systems are almost always better at performing their functions ("Keep It Simple Stupid"—KISS) (Beck 2000; Thomas 2025). This idea has been validated in real-world applications (Banker and Datar 1989; Ogheneovo et al. 2014). Following this notion, we establish our first design rule: *(1) The framework must be as simple as possible, while achieving its users' goals*. In our case the goal would be the creation of a toxic and challenging environment for experimentation. The violation of this simple rule could explain the under-performance of Concordia or the frustrations of Balog et al. (2024). A natural extension of this rule is *(2) When we do need to add complexity, this needs to be justified both epistemologically and quantitatively*. Indeed, each contribution of our methodology is evaluated before being adopted (§5.2). From our experiments, we also encountered a new limitation: *(3) Complexity is directly related to researcher bias*; each new feature necessarily follows our own expectations with how human discussions work. The work of Park et al. (2023a) managed to derive interesting insights, exactly because it did *not* tamper with the way LLM users interacted.

With regard to functionality, we posit that synthetic discussion methodologies need to at least implement the following components: *(1) Context management*—since LLMs are stateless, and need to be fed information as prompts, *(2) Turn-taking*—as LLMs are trained as chatbot assistants, and therefore can not decide *not* to speak, *(3) Instructions given to the LLMs*—which may need to be diversified in multi-participant discussions.

### 3.2 Our methodology

**Context Management** We assume that the $h$ most recent comments provide sufficient context for the LLM users, facilitators, and annotators; a technique that works well in the context of discussions (Pavlopoulos et al. 2020). While techniques such as dynamic summarization (Balog et al. 2024), LLM self-critique (Yu et al. 2024), or memory modules (Vezhnevets et al. 2023) exist, they result in greater computational cost and a less transparent and explainable system.

**Turn Taking** In online fora users often create "comment chains" following-up on responses to their previous comments. Thus, for each discussion turn we either allow the previous user to respond (with a $40\%$ probability), or select another random participant ($60\%$). This probability was selected experimentally; larger values tend to create "debate"-style discussions between only two or three participants, while lower values tend to create scenarios with minimal interaction between them.[1] A facilitator can respond after every comment, or stay silent by emitting an empty string.

**Instruction Prompting** We use a standard instruction prompt for the participants instructing them to respond to repeatedly toxic comments. This was a necessary measure to bypass the extreme agreeableness of LLMs (as seen in §5.2 and in literature—Park et al. (2023a); Anthis et al. (2025)) and is an example of the trade-off between complexity and research bias v.s. the need to acquire meaningful data. Facilitators are instructed to respond only when necessary.

Following the paradigm presented by Abdelnabi et al. (2024), we assign roles to non-facilitator participants (e.g.,

[1] It would be interesting to analyze quantitatively how discussions change when modifying this parameter, but this is outside of the scope of our work.

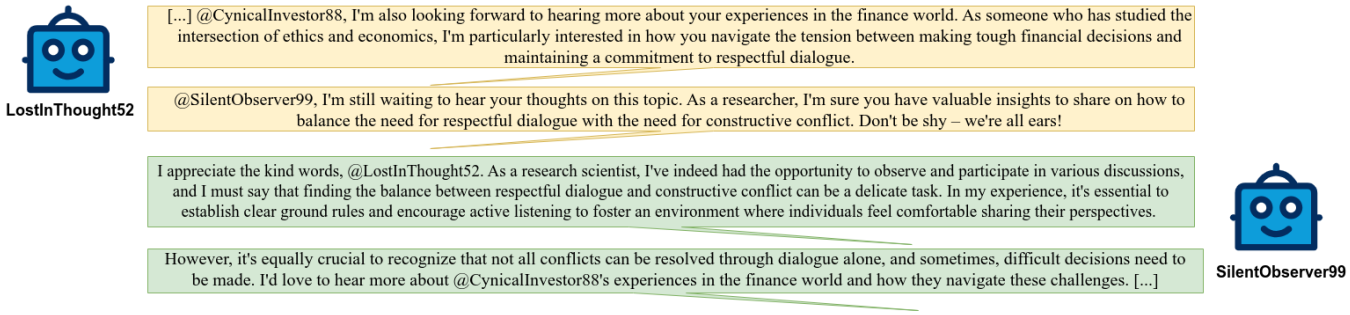



Figure 2: Excerpt from a synthetic discussion. The LLM participants use their sociodemographic prompts to insert personal stories and justify their perspectives in the discussion. They are also able to learn and retain information about other agents through conversation (a behavior also observed in Park et al. (2023a)). User `CynicalInvestor88` is also a part of the discussion; not a hallucination. Comments clipped due to length.

| Name | Type |
|---|---|
| Username | string |
| Age | integer |
| Gender | string |
| Education Level | string |
| Sexual Orientation | string |
| Demographic Group | string |
| Current Employment | string |
| Special Instructions | string |
| Personality Characteristics | list of strings |

Table 1: Sociodemographic information provided to the LLM participants and annotators. We defer the reader to the supplementary material for the actual values.

helping the community or disrupting discussions). We create three roles with distinct instructions for users (see supp. material): neutral users, trolls, and community veterans. We verify that roles have a quantitative effect on the toxicity of participants in §5.2.

**LLM Personas** Including SocioDemographic Backgrounds (SDBs—information such as gender, age and education) in prompts has proven promising in the generation of varied content and alleviation of Western bias exhibited by them (Burton et al. 2024). We generate 30 LLM user personas with unique SDBs (Table 1) by prompting a GPT-4 model (OpenAI 2024). Using these sociodemographic prompts, we observe that LLM users are able to create and share personal narratives and experiences from the provided information (Fig. 2).

# 4 Experimental Setup

## 4.1 Synthetic Discussion Generation

We use a set of starting comments ("seed opinions")—in our case controversial statements from Pavlopoulos and Likas (2024). We then run $N_d = 8$ discussions for each pair of facilitation strategies $S$ and LLM (§4.4). An overview of how the experiments are generated can be found in Algorithm 1. The *RandomSample* function returns a number of samples

Algorithm 1: Synthetic discussion setup generation

**Input:**
- User SDBs $\Theta = \{\theta_1, \ldots, \theta_{30}\}$
- Strategies $S = \{s_1, \ldots, s_6\}$
- Seed opinions $O = \{o_1, \ldots, o_7\}$
- LLMs = $\{LLaMa, Mistral, Qwen\}$

**Output:** Set of discussions $D$

$D = \{\}$
**for** $llm \in LLMs$ **do**
  **for** $s \in S$ **do**
    **for** $i = 1, 2, \ldots, N_d$ **do**
      $\hat{\Theta} = \textsc{RandomSample}(\Theta, num = 7)$
      $U = \textsc{Actors}(\text{llm}, \hat{\Theta})$
      $m = \textsc{Actors}(\text{llm}, s)$
      $o = \textsc{RandomSample}(O, num = 1)$
      $d = \{\text{users: } U, \text{ mod: } m, \text{ topic: } o\}$
      $D = D \cup d$
**return** $D$

from a set following the uniform distribution. The *Actors* function creates a LLM agent using a model and a prompt.

## 4.2 Facilitation Strategies

We test four different facilitation strategies, three of which are derived from Social Science research, along with two common-place strategies for discussion facilitation. Note that the process of turning sometimes extensive documents into short prompts, necessitated by open-source LLMs, is necessarily imperfect. We leave the optimal derivation of strategy prompts to future work.

1. **No Moderator**: A *common* strategy where no facilitator is present.
2. **No Instructions**: A *common* strategy where a LLM facilitator is present, but is provided only with basic instructions. This approach is already being used in some platforms (Tsai, Qian, and comm. contributors 2025). Example: "You are a moderator, keep the discussion civil".
3. **Rules Only**: A *real-life* strategy where the prompt is adapted from LLM alignment guidelines (Huang et al. 2024). These guidelines were selected to be as unani-

mously agreed upon as possible across various human groups. They thus provide a set of rules to uphold, without specifying *how* to uphold them, leaving the LLM completely unconstrained. Example: "Be fair and impartial, assist users, don't spread misinformation".

4. **Regulation Room**: A *real-life* strategy based on guidelines given to human facilitators of the "Regulation Room" platform (eRulemaking Initiative 2017). The instructions are suitable for online fora, where facilitators also engage in content moderation, and their effectiveness must be balanced by their throughput. Example: "Stick to a maximum of two questions, use simple and clear language, deal with off-topic comments".
5. **Constructive Communications**: A *real-life* strategy based on the human facilitation guidelines used by the MIT Center for Constructive Communications (White, Hunter, and Greaves 2024). It approaches facilitation from a more personalized and indirect angle, forbidding facilitators from directly providing opinions or directions. This makes the strategy ideal for deliberative environments. Example: "Do not make decisions, be a guide, provide explanations".
6. **Moderation Game**: Our proposed *experimental* strategy, inspired by Abdelnabi et al. (2024) (see §2.1). Instructions are formulated as a game, where the facilitator LLM tries to maximize their scores by arriving at specific outcomes. No actual score is being kept; they exist to act as indications for how desirable an outcome is. The other participants are not provided with scores, nor are they aware of the game rules. Example: "User is toxic: $-5$ points, User corrects behavior: $+10$ points".[2]

### 4.3 Evaluation

We use *toxicity* as a proxy for discussion quality, since it can inhibit online and deliberative discussions (De Kock, Stafford, and Vlachos 2022; Xia et al. 2020)[3]. We use ten LLM annotators controlled by the LLaMa3.1 70B model, which has been used in prior work (Kang and Qian 2024). LLMs are reliable for toxicity detection (Wang and Chang 2022; Anjum and Katarya 2024), thus allowing us to avoid problems of circular bias in our analysis.

In order to gauge the quality of our synthetic discussions, since we can not reliably measure "realism" (§2.4), we use the "diversity" metric (Ulmer et al. 2024). Low diversity points to pathological problems (e.g., LLMs repeating previous comments). Conversely, extremely high diversity may indicate a lack of interaction between participants. A discussion where participants engage with each other will naturally feature some lexical overlap, including common terms or paraphrasing points made by others. We compare the distribution of diversity scores for synthetic discussions with that measured on sampled human discussions. This allows us to estimate the extent to which synthetic discussions approximate real-world content variety and participant interaction.

We note again that these metrics are better interpreted as heuristics of actual discussion and synthetic data quality respectively. More research is needed w.r.t. reliable and generalizable quality metrics.

### 4.4 Technical Details

We use three instruction-tuned, open-source models: LLaMa3.1 (70B), Qwen2.5 (33B), Mistral Nemo (12B), quantized to 4 bits and run using a set seed (42). All the experiments were collectively completed within four weeks of computational time, using two Quadro RTX 6000 GPUs. The execution script is available in the project's repository. We provide the $h = 3$ last comments for context.

## 5 Results

### 5.1 Main findings

**Finding 1: LLM facilitators significantly improve synthetic discussions over time.** Unmoderated discussions tend to display significantly higher levels of toxicity (Fig. 3, Table 2). A linear regression analysis of toxicity over time ($Adj.R^2 = 0.413$) reveals that trolls exhibit intense toxicity—on average $1.3288$ points above neutral users and $1.3112$ above community veterans ($p < .000$) which decreases by an average of $-0.0125$ points per turn ($p = 0.003$). This trend is even more pronounced for neutral participants and community veterans, whose toxicity drops by $-0.0225$ ($p < .000$) and $-0.0350$ ($p < .000$) points per turn, respectively. This demonstrates the ability of the facilitator to reign in discussions over time, but also the diverging behaviors of different roles.

**Finding 2: Elaborate facilitation strategies fail to decrease toxicity.** The real-life strategies and our own strategy (§4.2) show a slight edge over time compared to *No Instructions*, but they do not consistently outperform it (Fig 3). This suggests that out-of-the-box LLMs may struggle to meaningfully incorporate complex instructions—which has been noted in prior work (Cho et al. 2024).

**Finding 3: LLM facilitators choose to intervene far too frequently, which is tolerated by the other participants.** Fig. 4 demonstrates that LLM facilitators intervene at almost any opportunity, even though they are instructed to only do so when necessary. This confirms that LLMs generally can not decide not to speak even when instructed to do so (§3.2). To our knowledge, this has not been reported in relevant literature, and *is an example of "debugging" problems with LLMs* — a core motivation of our work.

Additionally, we note that LLM participants exhibit atypical tolerance for excessive facilitator interventions. Humans in contrast typically become irritated and more toxic after repeated, unneeded interventions (Schaffner et al. 2024; Amaury and Stefano 2022; Schluger et al. 2022; Cresci, Trujillo, and Fagni 2022). This is likely another artifact of LLMs being too agreeable (Park et al. 2023a; Anthis et al. 2025).

[2] This could serve as a basis for a similar methodology based on game-theory, or as a Reinforcement Learning formulation for training. In this work we only explore whether the prompt itself can have an effect on the LLM facilitator; we leave the aforementioned approaches for future work.

[3] We note that this is not always true (Avalle et al. 2024).

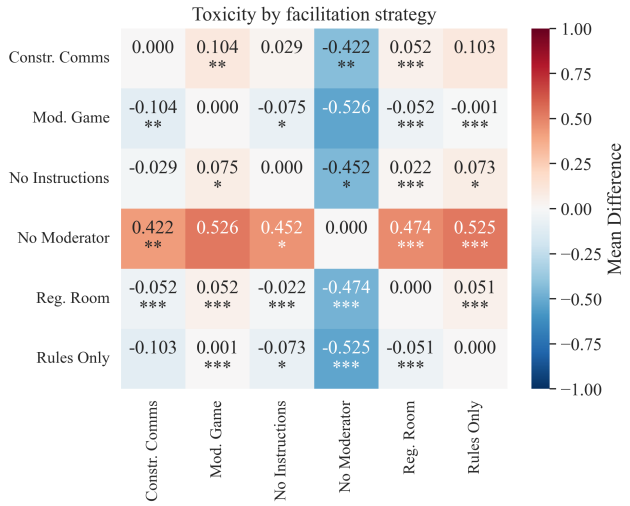



Figure 3: Difference in average toxicity levels for comments following pairs of facilitation strategies. Red cells ($x > 0$) indicate that the strategy on the left performs worse than the one on the bottom, for an average of $x$ points in a scale of 1-5. Conversely for blue ($x < 0$) cells. White cells denote minute changes. Asterisks derived from pairwise Student-t tests ( $\cdot p < 0.1$, $^{*}$ $p < 0.05$, $^{**}$ $p < 0.01$, $^{***}$ $p < 0.001$ ). The large size of our dataset allows using parametric tests.

## 5.2 Ablation Study

We generate eight synthetic discussions per ablation experiment, using a single model (Qwen 2.5). We compare the diversity (cf. §2.3, §4.3) of these discussions with our broader synthetic dataset, as well as the CeRI "Regulation Room" dataset.[4] We pick this dataset because it is publicly available and comprised of facilitated online human discussions on ten diverse topics.

**Each component of our methodology surpasses baselines in data quality.** We compare our turn-taking function (§3.2) to two baselines: Round Robin (participants speaking one after the other, then repeating) and Random Selection (uniformly sampling another participant each turn). Fig. 5b demonstrates that although all distributions diverge from the human distribution, our function is the only one not exhibiting extremely high diversity (i.e., very limited participant interaction §4.3). Fig. 5c illustrates that each individual prompting design decision (SDBs, roles, and instruction prompts) results in diversity scores more closely aligned with human discussions.

**Larger models do not increase the quality of discussions.** As shown in Fig. 5a, Qwen demonstrated the highest diversity among the evaluated models, indicating limited participant interaction (§2.3), followed by Mistral Nemo and LLaMa. It's worth noting that none of the models closely matched the diversity observed in human discussions.

[4]Disclaimer: Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the CeRI.

| Variable | Toxicity |
|---|---|
| Intercept | $2.164^{***}$ |
| No Instructions | $-0.426^{***}$ |
| Moderation Game | $-0.435^{***}$ |
| Rules Only | $-0.461^{***}$ |
| Regulation Room | $-0.277^{***}$ |
| Constructive Communications | $-0.230^{***}$ |
| time | $-0.012^{**}$ |
| No Instructions×time | -0.003 |
| Moderation Game×time | $-0.011^{*}$ |
| Rules Only×time | -0.008 |
| Regulation Room×time | $-0.023^{***}$ |
| Constructive Communications×time | $-0.023^{***}$ |

$\cdot p < 0.1$, $^{*}$ $p < 0.05$, $^{**}$ $p < 0.01$, $^{***}$ $p < 0.001$

Table 2: OLS regression coefficients for toxicity on the non-facilitator comments ($Adj.R^2 = 0.054$). Reference factor is *No Moderator*. All strategies outperform *No Moderator* in general. The *Regulation Room* and *Constructive Communications* real-life strategies additionally show improvements over time compared to *No Instructions*.

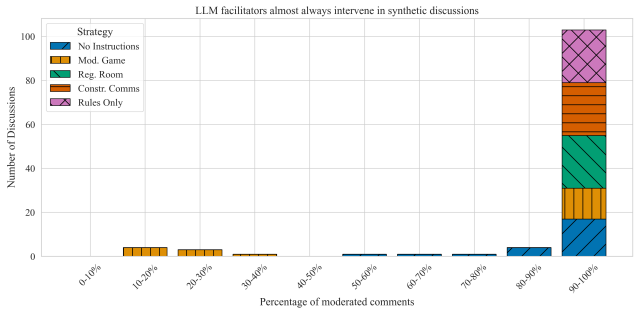



Figure 4: Histogram of interventions by LLM facilitators per strategy used.

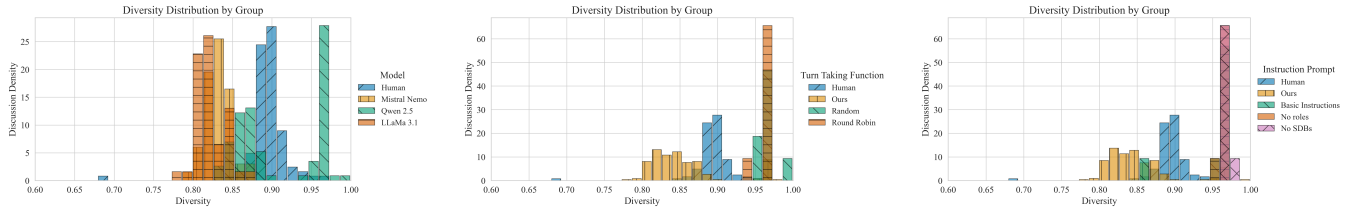
**Specialized instruction prompts are essential for eliciting toxic behavior in instruction-tuned LLMs.** Inserting trolls into the discussion leads to more intense toxicity among *other* participants *only if we instruct them to react to toxic posts* (Fig. 6).

# 6 Datasets and Software

We introduce an open-source, lightweight, purpose-built framework for managing, annotating, and generating synthetic discussions. The key features of the framework are:

- Three core functions: generating discussion setups (selecting participants, topics, roles, etc.), executing, and annotating them according to user-provided parameters.
- Built-in fault tolerance (automated recovery and intermittent saving) and file logging to support extended experiments.
- Availability via PIP.

We also release a dataset of synthetic discussions annotated by LLMs. It can serve for finetuning facilitator




(a) Model (b) Turn-taking function (c) Instruction prompt

Figure 5: Diversity (§2.3) distribution for each discussion by LLM (§4.4), turn-taking function $t$, and prompting function $\phi$ used (§3.2). Comparison with the CeRI Regulation Room dataset ("Human"). Note that the x-axis starts from 0.6.

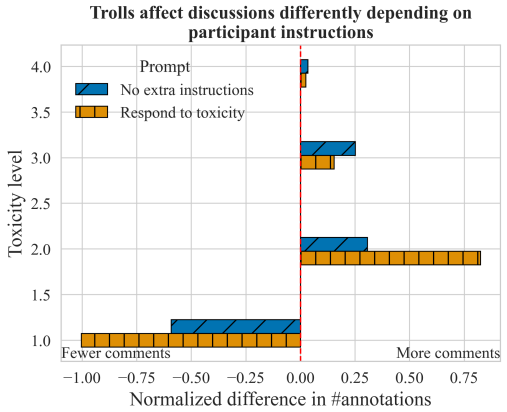



Figure 6: Non-troll toxicity levels in discussions with and without trolls. There is a significant uptick on the number of "somewhat toxic" ($Toxicity = 2$) comments when the participants are instructed to respond to toxic comments.

LLMs. We note that, as is the case with most synthetic datasets (Ulmer et al. 2024), the data may need to be filtered to derive only high-quality samples—in our case filtering out discussions with constant facilitator interventions or low/extremely high diversity. However, the data can be scaled accordingly, due to the low computational cost of our methodology. The supplementary ablation dataset, as well as the code for the analysis and the graphs present in this paper, can be found in the project repository. The dataset is licensed under a CC BY-SA license, and the software under GPLv3. **Warning: The datasets by their nature contain offensive and hateful speech.**

## 7 Conclusion

We conducted a brief overview of the trade-off between complexity and efficiency for synthetic discussion methodologies, from which we derived three simple design rules. Following these rules, we proposed a simple and generalizable methodology, whose components are easily validated and which enables researchers to quickly and inexpensively conduct pilot facilitation experiments using exclusively LLMs. We found that LLM facilitators significantly improve the quality of synthetic discussions; but prompting these facilitators with strategies based on Social Science research does not markedly improve their performance. We also discovered that LLM facilitators constantly intervene, even when instructed not to. Finally, we created an open-source Python framework which applies our methodology to hundreds of experiments. We used it to create and publish a large-scale synthetic dataset, which can be used for finetuning.

## 8 Discussion

**Future work** Future work should identify additional quality metrics to evaluate synthetic data, and discussion quality. The latter can then be used to examine the applicability of our findings obtained regarding optimal facilitation strategies, to discussions involving humans. It would also be interesting to explore how to more effectively prompt LLMs with complex facilitation strategies, or alternative formulations of our methodology, as described in this paper.

**Limitations** Given the limited prior research our methodology is mostly exploratory, and is evaluated with baselines using only two metrics. Our setup is restricted by the statelessness of LLMs, which forces us to overwhelmingly resort to prompting—however the use of open-source models prevented us from experimenting with more elaborate prompts requiring extended context windows.

**Ethical considerations** Synthetic discussions involving LLMs could be exploited by malicious actors to train them at performing unethical tasks (Majumdar et al. 2024; Marulli, Paganini, and Lancellotti 2024; Li et al. 2025), although ongoing research is addressing these vulnerabilities (Wang et al. 2025). Furthermore, the use of LLMs inherently risks skewing moderation systems towards the predominant demographics best represented in their training data. SDB prompts are a necessary but insufficient step towards avoiding this (Rossi, Harrison, and Shklovski 2024; Anthis et al. 2025; Burton et al. 2024).

**AI use statement** LLMs were used solely for styling and text corrections in this document. They were also partially used for generating framework documentation, and code for some of the presented graphs. All such changes and additions have been checked by the authors.

## References

Abdelnabi, S.; Gomaa, A.; Sivaprasad, S.; Schönherr, L.; and Fritz, M. 2024. Cooperation, Competition, and

Maliciousness: LLM-Stakeholders Interactive Negotiation. arXiv:2309.17234.

Abramski, K.; Citraro, S.; Lombardi, L.; Rossetti, G.; and Stella, M. 2023. Cognitive Network Science Reveals Bias in GPT-3, GPT-3.5 Turbo, and GPT-4 Mirroring Math Anxiety in High-School Students. *Big Data and Cognitive Computing*, 7(3).

Amaury, T.; and Stefano, C. 2022. Make Reddit Great Again: Assessing Community Effects of Moderation Interventions on r/The_Donald. *Proceedings of the ACM on Human-Computer Interaction*, 6: 1 – 28.

Anjum; and Katarya, R. 2024. Hate speech, toxicity detection in online social media: a recent survey of state of the art and opportunities. *International Journal of Information Security*, 23(1): 577–608.

Anthis, J. R.; Liu, R.; Richardson, S. M.; Kozlowski, A. C.; Koch, B.; Evans, J.; Brynjolfsson, E.; and Bernstein, M. 2025. LLM Social Simulations Are a Promising Research Method. arXiv:2504.02234.

Argyle, L. P.; Bail, C. A.; Busby, E. C.; Gubler, J. R.; Howe, T.; Rytting, C.; Sorensen, T.; and Wingate, D. 2023. Leveraging AI for democratic discourse: Chat interventions can improve online political conversations at scale. *Proceedings of the National Academy of Sciences*, 120(41): 1–8.

Atil, B.; Aykent, S.; Chittams, A.; Fu, L.; Passonneau, R. J.; Radcliffe, E.; Rajagopal, G. R.; Sloan, A.; Tudrej, T.; Ture, F.; Wu, Z.; Xu, L.; and Baldwin, B. 2025. Non-Determinism of "Deterministic" LLM Settings. arXiv:2408.04667.

Avalle, M.; Marco, N. D.; Etta, G.; Sangiorgio, E.; Alipour, S.; Bonetti, A.; Alvisi, L.; Scala, A.; Baronchelli, A.; Cinelli, M.; and Quattrociocchi, W. 2024. Persistent interaction patterns across social media platforms and over time. *Nature*, 628: 582 – 589.

Balog, K.; Palowitch, J.; Ikica, B.; Radlinski, F.; Alvari, H.; and Manshadi, M. 2024. Towards Realistic Synthetic User-Generated Content: A Scaffolding Approach to Generating Online Discussions. arXiv:2408.08379.

Banker, R. D.; and Datar, S. M. 1989. Software Complexity and Maintainability. In *Proceedings of the 10th International Conference on Information Systems (ICIS)*. Association for Information Systems, Boston, MA: AIS Electronic Library (AISeL).

Beck, K. 2000. *Extreme programming explained: embrace change*. addison-wesley professional.

Bisbee, J.; Clinton, J. D.; Dorff, C.; Kenkel, B.; and Larson, J. M. 2024. Synthetic Replacements for Human Survey Data? The Perils of Large Language Models. *Political Analysis*, 32(4): 401–416.

Burton, J. W.; Lopez-Lopez, E.; Hechtlinger, S.; et al. 2024. How Large Language Models Can Reshape Collective Intelligence. *Nature Human Behaviour*, 8: 1643–1655.

Chang, J. P.; and Danescu, C. 2019. Trouble on the Horizon: Forecasting the Derailment of Online Conversations as they Develop. In Inui, K.; Jiang, J.; Ng, V.; and Wan, X., eds., *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 4743–4754. Hong Kong, China: Association for Computational Linguistics.

Cho, H.; Liu, S.; Shi, T.; Jain, D.; Rizk, B.; Huang, Y.; Lu, Z.; Wen, N.; Gratch, J.; Ferrara, E.; and May, J. 2024. Can Language Model Moderators Improve the Health of Online Discourse? In Duh, K.; Gomez, H.; and Bethard, S., eds., *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, 7478–7496. Mexico City, Mexico.

Clemons, E. K.; Schreieck, M.; and Waran, R. V. 2025. Managing disinformation on social media platforms. *Electronic Markets*, 35(52).

Cresci, S.; Trujillo, A.; and Fagni, T. 2022. Personalized Interventions for Online Moderation. In *Proceedings of the 33rd ACM Conference on Hypertext and Social Media*, HT '22, 248–251. New York, NY, USA: Association for Computing Machinery. ISBN 9781450392334.

De Kock, C.; Stafford, T.; and Vlachos, A. 2022. How to disagree well: Investigating the dispute tactics used on Wikipedia. In Goldberg, Y.; Kozareva, Z.; and Zhang, Y., eds., *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, 3824–3837. Abu Dhabi, United Arab Emirates: Association for Computational Linguistics.

Denniss, E.; and Lindberg, R. 2025. Social media and the spread of misinformation: infectious and a threat to public health. *Health Promotion International*, 40.

eRulemaking Initiative, C. 2017. CeRI (Cornell e-Rulemaking) Moderator Protocol. Cornell e-Rulemaking Initiative Publications, 21.

Esau, K.; Friess, D.; and Eilders, C. 2017. Design matters! An empirical analysis of online deliberation on different news platforms. *Policy & Internet*, 9(3): 321–342.

Falk, N.; Jundi, I.; Vecchi, E. M.; and Lapesa, G. 2021. Predicting Moderation of Deliberative Arguments: Is Argument Quality the Key? In Al-Khatib, K.; Hou, Y.; and Stede, M., eds., *Proceedings of the 8th Workshop on Argument Mining*, 133–141. Punta Cana, Dominican Republic: Association for Computational Linguistics.

Falk, N.; Vecchi, E.; Jundi, I.; and Lapesa, G. 2024. Moderation in the Wild: Investigating User-Driven Moderation in Online Discussions. In Graham, Y.; and Purver, M., eds., *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, 992–1013. St. Julian's, Malta: Association for Computational Linguistics.

Gligori'c, K.; Zrnic, T.; Lee, C.; Candes, E. J.; and Jurafsky, D. 2024. Can Unconfident LLM Annotations Be Used for Confident Conclusions? *ArXiv*, abs/2408.15204.

Grossmann, I.; Feinberg, M.; Parker, D.; Christakis, N.; Tetlock, P.; and Cunningham, W. 2023. AI and the transformation of social science research. *Science (New York, N.Y.)*, 380: 1108–1109.

Habernal, I.; and Gurevych, I. 2016. Which argument is more convincing? Analyzing and predicting convincingness of Web arguments using bidirectional LSTM. In Erk, K.; and Smith, N. A., eds., *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 1589–1599. Berlin, Germany: Association for Computational Linguistics.

Hewitt, L.; Ashokkumar, A.; Ghezae, I.; and Willer, R. 2024. Predicting Results of Social Science Experiments Using Large Language Models. Equal contribution, order randomized.

Horta Ribeiro, M.; Cheng, J.; and West, R. 2023. Automated Content Moderation Increases Adherence to Community Guidelines. In *Proceedings of the ACM Web Conference 2023*, WWW '23, 2666–2676. New York, NY, USA: Association for Computing Machinery. ISBN 9781450394161.

Huang, S.; Siddarth, D.; Lovitt, L.; Liao, T. I.; Durmus, E.; Tamkin, A.; and Ganguli, D. 2024. Collective Constitutional AI: Aligning a Language Model with Public Input. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency*, FAccT '24, 1395–1417. New York, NY, USA: Association for Computing Machinery. ISBN 9798400704505.

Jaidka, K. 2022. Talking politics: Building data-driven lexica to measure political discussion quality. *Computational Communication Research*, 4(2): 486–527.

Jansen, B. J.; gyo Jung, S.; and Salminen, J. 2023. Employing large language models in survey research. *Natural Language Processing Journal*, 4: 100020.

Kang, H.; and Qian, T. 2024. Implanting LLM's Knowledge via Reading Comprehension Tree for Toxicity Detection. In Ku, L.-W.; Martins, A.; and Srikumar, V., eds., *Findings of the Association for Computational Linguistics ACL 2024*, 947–962. Bangkok, Thailand and virtual meeting: Association for Computational Linguistics.

Kim, S.; Eun, J.; Seering, J.; and Lee, J. 2021. Moderator Chatbot for Deliberative Discussion: Effects of Discussion Structure and Discussant Facilitation. *Proc. ACM Hum.-Comput. Interact.*, 5(CSCW1).

Kolluri, A.; Murthy, D.; and Vinton, K. 2025. Quantifying the spread of racist content on fringe social media: A case study of Parler. *Big Data & Society*, 12(2).

Korre, K.; Tsirmpas, D.; Gkoumas, N.; Cabalé, E.; Kontarinis, D.; Myrtzani, D.; Evgeniou, T.; Androutsopoulos, I.; and Pavlopoulos, J. 2025. Evaluation and Facilitation of Online Discussions in the LLM Era: A Survey. ACL ARR 2025 February Submission.

Kumar, D.; AbuHashem, Y. A.; and Durumeric, Z. 2024. Watch Your Language: Investigating Content Moderation with Large Language Models. *Proceedings of the International AAAI Conference on Web and Social Media*, 18(1): 865–878.

Leng, Y.; and Yuan, Y. 2024. Do LLM Agents Exhibit Social Behavior? arXiv:2312.15198.

Li, A.; Zhou, Y.; Raghuram, V. C.; Goldstein, T.; and Goldblum, M. 2025. Commercial LLM Agents Are Already Vulnerable to Simple Yet Dangerous Attacks. arXiv:2502.08586.

Lin, C.-Y. 2004. ROUGE: A Package for Automatic Evaluation of Summaries. In *Text Summarization Branches Out*, 74–81. Barcelona, Spain: Association for Computational Linguistics.

Lukito, J.; Chen, B.; Masullo, G. M.; and Stroud, N. J. 2024. Comparing a BERT Classifier and a GPT classifier for Detecting Connective Language Across Multiple Social Media. In Al-Onaizan, Y.; Bansal, M.; and Chen, Y.-N., eds., *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 19140–19153. Miami, Florida, USA: Association for Computational Linguistics.

Majumdar, D.; S, A.; Boyina, P.; Rayidi, S. S. P.; Sai, Y. R.; and Gangashetty, S. V. 2024. Beyond Text: Nefarious Actors Harnessing LLMs for Strategic Advantage. In *2024 International Conference on Intelligent Systems for Cybersecurity (ISCS)*, 1–7.

Marulli, F.; Paganini, P.; and Lancellotti, F. 2024. The Three Sides of the Moon LLMs in Cybersecurity: Guardians, Enablers and Targets. *Procedia Computer Science*, 246: 5340–5348. 28th International Conference on Knowledge Based and Intelligent information and Engineering Systems (KES 2024).

Marzo, G. D.; Pietronero, L.; and Garcia, D. 2023. Emergence of Scale-Free Networks in Social Interactions among Large Language Models. arXiv:2312.06619.

Mou, X.; Wei, Z.; and Huang, X. 2024. Unveiling the Truth and Facilitating Change: Towards Agent-based Large-scale Social Movement Simulation. arXiv:2402.16333.

Navajas, J.; Niella, T.; and Garbulsky, G. e. a. 2018. Aggregated knowledge from a small number of debates outperforms the wisdom of large crowds. *Nature Human Behaviour*, 2: 126–132.

Neumann, T.; De-Arteaga, M.; and Fazelpour, S. 2025. Should you use LLMs to simulate opinions? Quality checks for early-stage deliberation. arXiv:2504.08954.

Nik Ahmad, N. A. 2010. CETLs : Supporting Collaborative Activities Among Students and Teachers Through the Use of Think- Pair-Share Techniques. *International Journal of Computer Science Issues*, 7.

Ogheneovo, E. E.; et al. 2014. On the relationship between software complexity and maintenance costs. *Journal of Computer and Communications*, 2(14): 1.

OpenAI. 2024. GPT-4 Technical Report. arXiv:2303.08774.

Park, J. S.; O'Brien, J. C.; Cai, C. J.; Morris, M. R.; Liang, P.; and Bernstein, M. S. 2023a. Generative Agents: Interactive Simulacra of Human Behavior. arXiv:2304.03442.

Park, J. S.; O'Brien, J. C.; Cai, C. J.; Morris, M. R.; Liang, P.; and Bernstein, M. S. 2023b. Generative Agents: Interactive Simulacra of Human Behavior. *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*.

Park, J. S.; Popowski, L.; Cai, C.; Morris, M. R.; Liang, P.; and Bernstein, M. S. 2022. Social Simulacra: Creating Populated Prototypes for Social Computing Systems. In

*Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology*, UIST '22. New York, NY, USA: Association for Computing Machinery. ISBN 9781450393201.

Park, J. S.; Zou, C. Q.; Shaw, A.; Hill, B. M.; Cai, C.; Morris, M. R.; Willer, R.; Liang, P.; and Bernstein, M. S. 2024. Generative Agent Simulations of 1,000 People. arXiv:2411.10109.

Pavlopoulos, J.; and Likas, A. 2024. Polarized Opinion Detection Improves the Detection of Toxic Language. In Graham, Y.; and Purver, M., eds., *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, 1946–1958. St. Julian's, Malta: Association for Computational Linguistics.

Pavlopoulos, J.; Sorensen, J.; Dixon, L.; Thain, N.; and Androutsopoulos, I. 2020. Toxicity Detection: Does Context Really Matter? In Jurafsky, D.; Chai, J.; Schluter, N.; and Tetreault, J., eds., *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 4296–4305. Online: Association for Computational Linguistics.

Persing, I.; and Ng, V. 2015. Modeling Argument Strength in Student Essays. In Zong, C.; and Strube, M., eds., *Proceedings of the 53rd Annual Meeting of the Association for Computational Linguistics and the 7th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, 543–552. Beijing, China: Association for Computational Linguistics.

Pranesh, S.; and Gupta, S. 2024. The impact of social media on polarization in the society. arXiv:2408.12877.

Rosenberg, M. B.; and Chopra, D. 2015. *Nonviolent communication: A language of life: Life-changing tools for healthy relationships*. PuddleDancer Press.

Rossetti, G.; Stella, M.; Cazabet, R.; Abramski, K.; Cau, E.; Citraro, S.; Failla, A.; Improta, R.; Morini, V.; and Pansanella, V. 2024. Y Social: an LLM-powered Social Media Digital Twin. arXiv:2408.00818.

Rossi, L.; Harrison, K.; and Shklovski, I. 2024. The Problems of LLM-generated Data in Social Science Research. *Sociologica*, 18(2): 145–168.

Schaffner, B.; Bhagoji, A. N.; Cheng, S.; Mei, J.; Shen, J. L.; Wang, G.; Chetty, M.; Feamster, N.; Lakier, G.; and Tan, C. 2024. "Community Guidelines Make this the Best Party on the Internet": An In-Depth Study of Online Platforms' Content Moderation Policies. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, CHI '24. New York, NY, USA: Association for Computing Machinery. ISBN 9798400703300.

Schluger, C.; Chang, J.; Danescu-Niculescu-Mizil, C.; and Levy, K. 2022. Proactive Moderation of Online Discussions: Existing Practices and the Potential for Algorithmic Support. *Proc. ACM Hum.-Comput. Interact.*, 6(CSCW2).

Schroeder, H.; Roy, D.; and Kabbara, J. 2024. Fora: A corpus and framework for the study of facilitated dialogue. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 13985–14001. Bangkok, Thailand.

Seering, J. 2020. Reconsidering Self-Moderation: the Role of Research in Supporting Community-Based Models for Online Content Moderation. *Proc. ACM Hum.-Comput. Interact.*, 4(CSCW2).

Small, C. T.; Vendrov, I.; Durmus, E.; Homaei, H.; Barry, E.; Cornebise, J.; Suzman, T.; Ganguli, D.; and Megill, C. 2023. Opportunities and Risks of LLMs for Scalable Deliberation with Polis. *ArXiv*, abs/2306.11932.

Taubenfeld, A.; Dover, Y.; Reichart, R.; and Goldstein, A. 2024. Systematic Biases in LLM Simulations of Debates. *ArXiv*, abs/2402.04049.

Thomas, D. 2025. *simplicity: sustainable, humane, and effective software development*. The Pragmatic Programmers LLC.

Tsai, L. L.; Pentland, A.; Braley, A.; Chen, N.; Enríquez, J. R.; and Reuel, A. 2024. Generative AI for Pro-Democracy Platforms. *An MIT Exploration of Generative AI*. Https://mit-genai.pubpub.org/pub/mn45hexw.

Tsai, V.; Qian, C.; and comm. contributors, D. L. 2025. Deliberate Lab: Open-Source Platform for LLM-Powered Social Science.

Törnberg, P.; Valeeva, D.; Uitermark, J.; and Bail, C. 2023. Simulating Social Media Using Large Language Models to Evaluate Alternative News Feed Algorithms. arXiv:2310.05984.

Ulmer, D.; Mansimov, E.; Lin, K.; Sun, J.; Gao, X.; and Zhang, Y. 2024. Bootstrapping LLM-based Task-Oriented Dialogue Agents via Self-Talk. *ArXiv*, abs/2401.05033.

Vezhnevets, A. S.; Agapiou, J. P.; Aharon, A.; Ziv, R.; Matyas, J.; Du'enez-Guzm'an, E. A.; Cunningham, W. A.; Osindero, S.; Karmon, D.; and Leibo, J. Z. 2023. Generative agent-based modeling with actions grounded in physical, social, or digital space using Concordia. *ArXiv*, abs/2312.03664.

Wachsmuth, H.; Naderi, N.; Hou, Y.; Bilu, Y.; Prabhakaran, V.; Thijm, T. A.; Hirst, G.; and Stein, B. 2017. Computational Argumentation Quality Assessment in Natural Language. In Lapata, M.; Blunsom, P.; and Koller, A., eds., *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 1, Long Papers*, 176–187. Valencia, Spain: Association for Computational Linguistics.

Wang, H.; Fu, W.; Tang, Y.; Chen, Z.; Huang, Y.; Piao, J.; Gao, C.; Xu, F.; Jiang, T.; and Li, Y. 2025. A Survey on Responsible LLMs: Inherent Risk, Malicious Use, and Mitigation Strategy. arXiv:2501.09431.

Wang, Y.-S.; and Chang, Y. T. 2022. Toxicity Detection with Generative Prompt-based Inference. *ArXiv*, abs/2205.12390.

White, K.; Hunter, N.; and Greaves, K. 2024. *facilitating deliberation - a practical guide*. Mosaic Lab.

Xia, Y.; Zhu, H.; Lu, T.; Zhang, P.; and Gu, N. 2020. Exploring Antecedents and Consequences of Toxicity in Online Discussions: A Case Study on Reddit. *Proc. ACM Hum.-Comput. Interact.*, 4(CSCW2).

Yu, Y.; Yao, Z.; Li, H.; Deng, Z.; Cao, Y.; Chen, Z.; Suchow, J. W.; Liu, R.; Cui, Z.; Xu, Z.; Zhang, D.; Subbalakshmi, K.; Xiong, G.; He, Y.; Huang, J.; Li, D.; and Xie, Q. 2024. FinCon: A Synthesized LLM Multi-Agent System with Conceptual Verbal Reinforcement for Enhanced Financial Decision Making. arXiv:2407.06567.

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4.7. If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results (yes/partial/no/NA) yes

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